

Kohtaro Yamakawa

CONTACT INFORMATION	Berkeley, CA +1 (614) 649-5553 kohyamakawa@gmail.com	
RESEARCH INTERESTS	synthesis, electric transport, materials discovery, topology, magnetic semiconductors, magnetically frustrated systems, superconductivity & correlated electrons	
EDUCATION	University of California, Berkeley , Berkeley, CA Ph.D. in Physics Adviser: Prof. James G. Analytis	August 2026
	Columbia University , New York, NY B.A. in Physics, B.A. in Math	May 2020
RESEARCH APPOINTMENTS	University of California, Berkeley , Department of Physics Adviser: James G. Analytis Single crystal growth, discovery, and characterization of topological quantum materials.	August 2020 - August 2026
	Columbia University , Department of Physics Adviser: Yasutomo J. Uemura μ SR spectroscopy and analysis of strongly correlated systems including iron-based superconductors and Mott Insulators.	June 2017-August 2020
	Columbia University , Department of Mathematics Adviser: Jacob Shapiro Analytical study of the bulk edge correspondence in 2D systems using Fredholm Theory.	Summer 2018
FUNDING AWARDS	Kavli ENSI Graduate Fellow Erwin Hahn Graduate Fellow Anonymous Internship Summer Fund Class of 1939 Summer Research Fellow Alumni Summer Internship Fellow	2025-2026 2021-2022 Summer 2019 Summer 2018 Summer 2017
OTHER HONORS AND AWARDS	Kavli Discovery Hub Fellow , <i>University of California, Berkeley</i> Received funding to lead independent research programs and engaged in workshops on becoming an independent investigator. Outstanding GSI Award , <i>University of California, Berkeley</i> Awarded to outstanding graduate student instructors.	2025 2021
MANUSCRIPTS (CHRONOLOGICAL)	K. Yamakawa , H. Pentz, L. Rouquette, K. Deoskar, and J.G. Analytis. Data-driven materials discovery of kagome materials toward observable flat band physics. In preparation. K. Yamakawa* , Q.Zhang*, C. Li, R. Reczek, R.P. Day, P. Fairfield, M. Allen, and J.G. Analytis. Quantum Oscillations in 2D van der waals magnet EuSn_2As_2 . In preparation L. Pritchard Cairns, K. Yamakawa , S. Zhang, Y. Chen, B. Field, R. Reczek, R.P. Day, J.E. Moore, M. Jaime, S. Griffin, R.J. Birgeneau, and J.G. Analytis. Colossal Magnetoresistance and Phonon Driven Dynamically Modified Exchange in $\text{Eu}_5\text{Sn}_2\text{As}_6$. arxiv:2605.06649	

H. Taghinejad, **K. Yamakawa**, X. Huang, Y. Lyu, L. Pritchard Cairns, S. Husain, R. Ramesh, and J.G. Analytis. Low-field regime of magnon transport in PLD-grown YIG films. *Nano Lett.* **25**, 16, 6438–6444 (2025)

R. Saatjian, S. Dovrén, **K. Yamakawa**, R.S. Russell, J.G. Analytis, J.W. Harter. Quantum decoherence by magnetic fluctuations in a magnetic topological insulator. *npj Quantum Mater.* **10**, 81 (2025)

R. P. Day*, **K. Yamakawa***, L. Pritchard Cairns, J. Singleton, W. L. Cao, C. Wu, M. Allen, J. E. Moore, and James G. Analytis. Colossal magnetoresistance in antiferromagnetic semiconductor $\text{Eu}_5\text{Sn}_2\text{As}_6$. *Phys. Rev. B* **111** 054406 (2025)

E. Donoway, T.V. Trevisan, A. Liebman-Peláez, R.P. Day, **K. Yamakawa**, Y. Sun, J.R. Soh, D. Prabhakran, A.T. Boothroyd, R.M. Fernandes, J.G. Analytis, J.E. Moore, J. Orenstein, and V. Sunko. Multimodal approach reveals the symmetry-breaking pathway to the broken helix in EuIn_2As_2 . *Phys. Rev. X* **14** 031013 (2025)

S. Toyoda, J. Kruppe, **K. Yamakawa**, J. G. Analytis, and J. Orenstein. Phase control of spin waves in the van der Waals antiferromagnet NiPS_3 *Phys. Rev. B* **109** 064408 (2024)

K. Matsuura, M. Roppongi, M. Qiu, Q. Sheng, Y. Cai, **K. Yamakawa**, Z. Guguchia, R.P. Day, K.M. Kojima, A. Damascelli, Y. Sugimura, M. Saito, T. Takenaka, K. Ishihara, Y. Mizukami, K. Hashimoto, Y. Gu, S. Guo, L. Fu, Z. Zhang, F. Ning, G. Zhao, G. Dai, C. Jin, J.W. Beare, G.M. Luke, Y.J. Uemura, and T. Shibauchi. Two superconducting states with broken time-reversal symmetry in $\text{FeSe}_{1-x}\text{S}_x$. *PNAS* **120** (21) e2208276120 (2023)

S. Hameed, S. El-Khatib, K.P. Olson, B. Yu, T.J. Williams, T. Hong, Q. Sheng, **K. Yamakawa**, J. Zang, Y.J. Uemura, G.Q. Zhao, C.Q. Jin, L. Fu, Y. Gu, F. Ning, Y. Cai, K. M. Kojima, J. W. Freeland, M. Matsuda, C. Leighton, and M. Greven. Nature of the ferromagnetic-antiferromagnetic transition in $\text{Y}_{1-x}\text{La}_x\text{TiO}_3$. *Phys. Rev. B* **104** 024410 (2021)

E. Fonseca, J. Shapiro, A. Sheta, A. Wang, and **K. Yamakawa**. Two-dimensional time-reversal-invariant topological insulators via Fredholm theory. *Math Phys Anal Geom* **23**: 29 (2020)

Z. Guguchia, B.A. Frandsen, D. Santos-Cottin, S.C. Cheung, Z. Gong, Q. Sheng, **K. Yamakawa**, A.M. Hallas, M.N. Wilson, Y. Cai, J. Beare, R. Khasanov, R. De Renzi, G.M. Luke, S. Shamoto, A. Gauzzi, Y. Klein, and Y.J. Uemura. Probing the quantum phase transition in Mott insulator BaCoS_2 tuned by pressure and Ni substitution. *Phys. Rev. Materials* **3**, 045001 (2019)

S.C. Cheung, Z. Guguchia, B.A. Frandsen, Z. Gong, **K. Yamakawa**, D.E. Almeida, I.J. Onuorah, P. Bonfá, E. Miranda, W. Wang, D.W. Tam, Y. Song, C. Cao, Y. Cai, A.M. Hallas, M.N. Wilson, T.J. S. Munsie, G. Luke, B. Chen, G. Dai, C. Jin, S. Guo, F. Ning, R.M. Fernandes, R. De Renzi, P. Dai, and Y.J. Uemura. Disentangling superconducting and magnetic orders in $\text{NaFe}_{1-x}\text{Ni}_x\text{As}$ using muon spin rotation. *Phys. Rev. B* **97**, 224508 (2018)

S. Guo, H. Man, K. Wang, C. Ding, Y. Zhao, L. Fu, Y. Gu, G. Zhi, B.A. Frandsen, S.C. Cheung, Z. Guguchia, **K. Yamakawa**, B. Chen, H. Wang, Z. Deng, C.Q. Jin, Y.J. Uemura, and F. Ning. $\text{Ba}(\text{Zn},\text{Co})_2\text{As}_2$: A diluted ferromagnetic semiconductor with n -type carriers and isostructural to 122 iron-based superconductors. *Phys. Rev. B* **99**, 155201 (2017)

*denotes co-first author with equal contribution.

PRESENTATIONS	<i>*Material Insights into the Discovery of Quantum Phenomena,</i>	September 2025
	Department Seminar, Max Planck for the Structure and Dynamics of Matter, Max Planck Chemical Physics of Solids, Columbia University, University of British Columbia	
	<i>*Superconductivity in Two Dimensions,</i>	May 2025
	Gordon Research Conference on Superconductivity, Les Diableret, Geneva	
	<i>Promise of Topology in Axionic Insulator Candidate Eu₅Sn₂As₆,</i>	September 2023
	PHYS 290K Condensed Matter Seminar Series, UC Berkeley	
	<i>*Colossal Magnetoresistance and Hierarchy of Competing orders in Eu₅Sn₂As₆</i>	March 2022
	APS March Meeting, Las Vegas, NV	
	<i>*Characterization of Axionic Insulator Candidate EuIn₂As₂</i>	March 2021
	APS March Meeting, Chicago, IL	
	<i>*Bulk Edge Correspondence of Topological Insulators using Fredholm Theory</i>	August 2019
	Department of Mathematics Seminar, Columbia University	

*denotes invited talk.

RESEARCH MENTORING	Undergraduate students:
	Keshav Deoskar, 2025-2026
	Kenzo Ota, 2024-2025
	Lex Rouquette, 2023-2025
	Parker Fairfield, 2023-2024
	Rainer Reczek, 2022-2025
	Salvador Rangel Huerta, 2022-2024
	Shimeng Yang, 2022-2023

TEACHING	Graduate Student Instructor, <i>University of California, Berkeley</i>	2020-2021
	Led discussion sections & office hours for Physics 7B: Electricity and Magnetism and PHYS 141A: Introduction to Solid State Physics.	
	Teaching Assistant, <i>Columbia University</i>	Spring 2020
	Graded problem sets for Physics 3007 Electricity and Magnetism. Organized discussions and facilitated transition to online synchronous teaching.	
	Mathematics Tutor, <i>Columbia Tutoring and Learning Center</i>	2017- 2018
	Taught mathematics to high school students at Columbia Secondary High School.	
	Student Teacher, <i>Splash at Columbia, MIT</i>	2016 - 2020
	Developed and taught unique courses for New York, New Jersey, and Connecticut based middle and high school students.	

SELECTED ACADEMIC SERVICE	Pi² Committee, <i>University of California, Berkeley</i>	2022 - 2025
	Led the committee tasked with selecting undergraduates for the Physics Innovators Initiative (Pi ²) summer research fellowship. Arranged community events and coordinated technical writing and introduction to python workshops.	
	Outreach Lectures, <i>Oakland</i>	November 2024
	Lectured and demonstrated superconductivity to high school students at Oakland Technical	

High School. Hosted a demonstration of the Meissner effect.

Co-organizer, QI Winter School, *Lawrence Berkeley National Laboratory* January 2023
Co-organized 2023 Munich-Berkeley Quantum Information Science Winter School inviting students and faculty from Technical University of Munich. Curated and invited speakers.

Co-organizer, The COMPASS Project, *University of California, Berkeley* 2022 - 2025
Coordinated events centered around creating an inclusive environment in the physics community. Facilitated a lecture series geared towards showcasing the diversity of physics research at Berkeley for undergraduates.

Admitted students Open House, *University of California, Berkeley* 2022 - 2024
Assisted in planning open house for students admitted to the graduate physics program. Coordinated tours, engaged in individual student recruitment.

Co-organizer, CDM Journal Club, *University of California, Berkeley* 2020 - 2024
Co-organized weekly condensed matter journal clubs for students and postdocs.

TECHNICAL SKILLS *applications:* python, bash, mathematica, julia
synthesis: metallic flux growth, chemical vapor transport, salt flux growth, Czochralski method, solid state growth
characterization techniques: X-ray diffraction, energy-dispersive X-ray spectroscopy, Laue diffraction, Raman spectroscopy
measurement techniques: electron transport, magnetization measurements, heat capacity, high magnetic field, μ SR spectroscopy
fabrication techniques: focused ion beam (FIB) patterning of Hall devices

PROFESSIONAL REFERENCES Prof. James G. Analytis
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